SunMint Program — Consolidated Progress Report (v4)

Prepared by: Sophia Truesight (autopilot) · **For:** Gary Teh & TrueSight DAO members · **Date:** 2026-08-26

Revision note: v4 adds the **ACORN/PM002 methodology with the actual carbon formula**, the **lean MRV stack decision** (phone primary, free satellite, drone deferred), and a **clear statement of the resource gaps** needed to execute.

1. The model — mobile-first, decentralized

SunMint Farmer App (Android + iOS) — `TrueSightDAO/sunmint_mobile`, Capacitor 8, 21+ PRs merged, UAT'd 2026-08-23. Offline-first, RSA-signed, PT/EN.

Evolution: web app → native mobile → **next: tree-growth monitoring in the app** — the phone becomes the farmer's carbon-MRV device.

2. Certification route (decided)

- **Pilot** → **Plan Vivo (PVCs)** under **PM002** — the Acorn methodology (active 29/09/2025): participatory farmer monitoring, lower validation costs, ≥60% of revenue to communities
- **Scale** → **Verra VM0047 (ARR) + CCB** — ICVCM-approved; corporate offtake path
- VM0017 corrected (SALM ≠ ARR) — PR #294

3. Landmark precedents

- **Andean Cacao (Colombia)** — first large-scale cacao agroforestry under Verra: VM0047 + CCB, Terra Global, **56,000+ VCUs first issuance**
- **ACORN (Rabobank + Solidaridad + Microsoft + Satelligence)** — cocoa/coffee agroforestry, Plan Vivo-certified (PM002), **~80% of credit value to farmers**, 4–6 tCO■e/ha/yr
- **TREEO** — calibration-card DBH via phone camera, 94–95% accuracy ($R^2 \geq 0.95$)
- **Greenstand Treetracker** — geotagged repeat photos, 500k+ trees, open-source
- **CommuniTree (Taking Root)** — 28M trees, Plan Vivo since 2010

4. THE METHODOLOGY — ACORN / Plan Vivo PM002 (the actual formula)

Source: Plan Vivo Climate Methodology **PM002 v1.0** — "Methodology for Quantifying Carbon Benefits from Small-Scale Agroforestry" (developer: Acorn; reviewers: Plan Vivo TRP & AENOR; active 29/09/2025). Public PDF: `planvivo.org` → PV Climate → Approved Methodologies → PM002.

The carbon calculation chain (the algorithm)

Step 1 — Belowground biomass from aboveground (Equation 1):

```
ΔBGB_p = ΔAGB_p × R
```

- ΔBGB_p = change in belowground biomass in measurement period p (tonnes)
- ΔAGB_p = change in aboveground biomass in period p (tonnes; per module PU006)
- R = root:shoot ratio — from IPCC (2019) Table 4.4 by ecological zone; default **0.32** (Kim, Kirschbaum & Beedy, 2016) if no zone can be mapped

Step 2 — Reported Plan Vivo Certificates (Equation 6.1):

```
rPVCs = ((ΔAGB_p + ΔBGB_p) × CF) × F × (1 – A_pre) × (1 – A_unc) × (1 – LD_CE) × (1 – AR) × (1 – RB) – (E_proj,p – E_base,p)
```

Step 3 — Verified Plan Vivo Certificates (Equation 6.2):

```
vPVCs = ((ΔAGB_p + ΔBGB_p) × CF) × F × (1 – A_pre) × (1 – A_unc) × (1 – LD_CE) × (1 – RB) – (E_proj,p – E_base,p)
```

Where (the constants that make it concrete):

Symbol	Meaning	Value
CF	Carbon fraction of biomass	**0.47**
F	C → CO■ conversion	**44/12 = 3.667**
A_pre	Pre-project tree biomass adjustment	% (module PU007)
A_unc	Uncertainty adjustment	% (module PU008)
LD_CE	Leakage discount (carbon pools & emission sources)	% (module PU004)
AR	Achievement reserve (withheld vs underperformance)	**10%**
RB	Contribution to future Risk Buffer	**20%**
E_proj,p – E_base,p	Project emissions minus baseline emissions	t CO■e

Step 0 — the tree measurement itself (phone-based, PU006):

- **DBH** measured at 1.3 m via calibration-card photo (TREEO method) or tape
- **Species** recorded → species-specific **allometric equation** → AGB (kg)

- Agroforestry allometric equations differ from forest equations (lower density, faster growth) — must be developed/validated per species. Example agroforestry equations (southern Ontario intercropping): Red Oak `0.00776·(D²H)^1.0856`; Black Walnut `0.0126·D^2.8417`; Black Locust `0.0132·D^2.6762` — where D = DBH (cm), H = height (m). For SunMint we need equations for **cacao, Brazil nut, açaí, mahogany, jatobá** (peer-reviewed or locally calibrated).
- Carbon stock = biomass × 0.5 (IPCC default) or 0.47 (PM002)

What this means for us:

The formula is **fully implementable in our mobile app** — per-tree DBH → allometric AGB → ΔAGB over time → Equation 6.1/6.2 → PVCs. This is the engine the `monitor_tree_growth` module will run.

5. THE LEAN MRV STACK (decision from the deep dive)

Layer	Role	Cost	Verdict
Farmer phone	PRIMARY — per-tree DBH/growth (TREEO method)	Low (farmer-owned)	**■ CORE — keep**
Satellite (Sentinel/Landsat)	Baseline proof (non-forest ≥10 yr) + leakage context	**$0 (free)**	**■ Keep, free tier**
Satellite AI (NOR Space)	Premium analysis, auto-PDD	$$$	**■ Optional — only if VVB demands**
Drone (PODream)	Calibration sampling	$5–50/ha	**■ DEFER — post-pilot, 5–10% of plots**

Why: ACORN itself is moving ground-truth to phone (Agerpoint 3D app replaces tape measures); EARSC confirms young agroforestry has weak remote-sensing signals — the tree is best measured on the ground; drones cost ~100× satellite per hectare.

VM0047 caveat: if cacao density >50 planting units/ha (it will be ~500–1000/ha), the project counts as land-use change → area-based approach → remote-sensing baseline becomes mandatory. Under **Plan Vivo PM002 pilot route**, participatory farmer monitoring is the standard — softer satellite requirement.

6. ■■ RESOURCE GAPS — what we need to execute (explicit)

People / capability

Gap	Need	Est. cost
PDD/PM002 document writer	Carbon methodology consultant (agroforestry)	$8–15k

Gap	Need	Est. cost
VVB engagement	Validation & verification body (Plan Vivo TRP-approved, e.g. AENOR, Earthood)	$10–20k
Field enumerator training	Train co-op farmers in calibration-card method	$3–6k
Allometric equation development	Local species equations (cacao, Brazil nut, açaí, mahogany, jatobá)	$5–10k (research/calibration)

Data

Gap	Need	Source
Baseline land-use proof	Non-forest ≥10 yr evidence	FREE Sentinel/Landsat archive analysis (GIS work)
Plot boundaries (GIS)	Parcel polygons per co-op	Survey + CAR records
CAR tenure docs	Land registration numbers	Co-ops
Community engagement records	FPIC / meeting minutes	Co-ops + us

Technology

Gap	Need	Effort
`monitor_tree_growth` module	Calibration-card photo → DBH → allometric CO■e → signed event	2–3 wks dev
Tree registry	Per-tree IDs in ledger	1–2 wks dev
Allometric engine	Species formula table (5 species)	1 wk dev
Satellite baseline ingestion	Free imagery processing pipeline	2–4 wks

Financial (revised — lean stack)

Item	Est.
Stage 0 — legal/tenure + baseline GIS + community records	**$5–10k** (was $10–15k with NOR)
Stage 1 — PDD + VVB + calibration	**$35–60k**
Stage 2 — pilot planting 20–50 ha	**$150–250k**
Stage 3 — scale to 10,000 ha	**$3–10M+**

The single biggest gap

Baseline data + PDD validation — nothing else moves until we can prove the land story (free satellite archive) and get the PDD written + VVB-attested. That is the Stage 0–1 unlock.

7. Prioritized execution path (revised for lean stack)

1. **Merge PR (lean stack + PM002)** — done this cycle
2. **Stage 0 (~$5–10k DAO injection):** legal/tenure, free-satellite baseline GIS, community records
3. **Build `monitor_tree_growth` module** in the mobile app (implements PM002 equation chain)
4. **Shortlist VVBs** (AENOR/Earthood — Plan Vivo TRP reviewers)
5. **Develop local allometric equations** for the 5 species
6. **Submit Terra RFP** once baseline + boundaries + tenure are documented
7. **Optional:** NOR Space quote ONLY if VVB requires premium analysis; PODream drone calibration post-pilot

Prepared by Sophia Truesight, autopilot · TrueSight DAO · Mission: restore 10,000 hectares of Amazon rainforest